

LETTERS

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Jennifer Sills



Bolivia has recently enacted decrees that put the jaguar (*Panthera onca*) and its habitat at risk.

Bolivia must prioritize biodiversity

In April, Bolivia's Agro-Environmental Court handed down a landmark ruling in favor of conserving the jaguar (*Panthera onca*) and its habitat by protecting environmental defenders, mitigating human-wildlife conflicts, creating a conservation fund, curbing illegal wildlife trafficking, and pausing burning and extractive activities in protected areas (1). However, in the weeks that followed, the Bolivian government enacted supreme decrees 5381 and 5390, which contradict the court's decision by allowing soybean production in the departments of La Paz and Beni and the management of forest reserves, which will allow for an increase in human activities (2). These measures will substantially increase land use change in jaguar habitat areas. Bolivia should revisit these decisions and prioritize biodiversity conservation, especially for species of conservation interest, such as the jaguar, which is classified as Near Threatened globally (3) but is considered Vulnerable in Bolivia (4).

With more than 48,000 registered species (and 79,000 estimated species), of which 12,245 are plants and 3296 are vertebrates (5), Bolivia is one of the most diverse countries on the planet (6). The country is home to 774 endemic species; 198 Near Threatened species; 655 species classified as Vulnerable, Endangered, or Critically Endangered; 1 Extinct species; and 4 species that are Extinct in the Wild (3). However, the destruction of the jaguars' habitat (7), illegal hunting (8) and the trafficking of body parts such as skins and teeth for illegal trade (9) are likely decimating Bolivia's jaguar population. Deforestation rates are second only to those of Brazil (10).

Despite the country's declining biodiversity, Bolivia allocates only 0.04% of the general state budget (a sum equivalent to US\$1.7 million) to conservation (11). Without urgent action to address biodiversity loss, Bolivia is heading toward further endangered species loss and potential ecosystem collapse. To reverse this trend, the next Bolivian government must increase the national conservation budget and create a government agency dedicated to promoting scientific research and actions to protect biological diversity. In addition, the next administration should strengthen the operational capacity and management efficiency of national and subnational protected areas and should improve land use planning

by prioritizing ecosystems and Key Biodiversity Areas (12). Bolivia can take immediate action by enforcing existing legal regulations that prioritize the biodiversity conservation and by honoring the Agro-Environmental Court's decision.

Oswaldo Maillard

Fundación para la Conservación del Bosque Chiquitano, Santa Cruz de la Sierra, Bolivia.
Email: omaillard@fcbc.org.bo

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3. Go to IUCN Red List of Threatened Species (2025) at <https://www.iucnredlist.org>. To find the jaguar's global conservation status, search for "*Panthera onca*." To find the list of species in Bolivia, select the "Advanced" option in the search bar. In the search filters menu on the left, expand the "Land Regions" category by clicking the arrow to the right of the text. Then expand the "South America" category. Select "Bolivia, Plurinational States of" by clicking the box to the left. Then expand the "Red List Category" option in the main filters list. The number of species in Bolivia in each category will display in parentheses. To find the Endemic category, scroll to the bottom of the filter list.
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Safe food: A human right amid climate change

Unsafe food leads to illness in an estimated 600 million people and causes 420,000 deaths globally each year (1, 2). Foodborne illness and death disproportionately affect children under the age of 5 and people in low-income countries (1). Access to safe and nutritious food is included in the United Nations (UN) Sustainable Development Goals as target 2.1, but food safety is often overlooked in many national adaptation plans (3, 4). The right to adequate food is globally recognized, but access to safe food is not explicitly guaranteed (5, 6). Given that access to safe food is critical to equitable and sustainable food systems and that climate change will exacerbate the likelihood of foodborne illnesses and deaths (2), the global community must explicitly enshrine food safety as a universal human right.

Global food safety efforts are primarily based on two frameworks: the World Health Organization's Global Strategy for Food Safety and the Codex Alimentarius, a set of voluntary guidelines enforceable only in international trade disputes (7–9). Many nations still lack the infrastructure and resources needed to implement these strategies and standards (8, 10). Moreover, these frameworks include neither dedicated funding mechanisms for implementing food safety nor binding enforcement mechanisms to ensure adherence to guidelines.

Public health challenges related to food safety will worsen as climate change increases safety hazards across terrestrial and aquatic food systems (2, 11). For example, warmer, more humid conditions promote aflatoxin contamination in sub-Saharan crops (12). Rising seawater temperatures increase harmful algal bloom risks in seafood from the Caribbean Sea and the Pacific and Indian Oceans (11). Extreme weather disrupts cold storage and transport, increasing the risk of food spoilage worldwide (11). Low- and middle-income countries, which already lose US\$110 billion annually to unsafe food, will face the greatest risks (10).

Although existing efforts have improved access to safe food, international organizations must take additional action to fully realize food safety for global communities. At the 80th meeting of the UN General Assembly in September, UN member states should initiate a resolution that affirms the human right to food safety. The resolution should mandate global cooperation to strengthen food safety systems, annual reporting on foodborne disease burdens and climate-related risks, and the integration of foodborne illness into national health surveillance systems. It should also mandate that UN members provide technical and financial support for low- and middle-income countries to build capacity for food safety surveillance and cold storage and transport, and it should require the establishment of global rapid response mechanisms to address food safety issues driven by climate change.

Christopher J. Knight¹, Theresa L. U. Burnham², Larry B. Crowder¹, Jacob G. Eurich^{3,4}, Nicole Franz⁵, Christopher D. Golden^{6,7}, Fiorenza Micheli^{1,5,8}

¹Hopkins Marine Station, Stanford University, Pacific Grove, CA, USA. ²School of Marine Sciences, University of Maine, Orono, ME, USA. ³Environmental Defense Fund, Santa Barbara, CA, USA. ⁴Marine Sciences Institute, University of California Santa Barbara, Santa Barbara, CA, USA. ⁵Stanford Center for Ocean Solutions, Stanford University, Stanford, CA, USA. ⁶Department of Environmental Health, Harvard T.H. Chan School of Public Health, Boston, MA, USA. ⁷Department of Nutrition, Harvard T.H. Chan School of Public Health, Boston, MA, USA. ⁸Oceans Department, Stanford University, Pacific Grove, CA, USA. Email: cjknight@stanford.edu

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Include scorpions in global conservation plans

Invertebrates are undergoing a rapid and mostly undocumented decline (1). Among the oldest terrestrial organisms (2), scorpions survived all five mass extinctions, but contemporary threats are testing their resilience. Most of the nearly 2900 scorpion species are ecologically specialized and geographically restricted, and some are confined to a single island, cave, or mountaintop (3). Anthropogenic pressures threaten many scorpion species, yet conservation efforts



Overharvesting and other human activities threaten thousands of scorpion species, including the emperor scorpion (*Pandinus imperator*).

have largely overlooked them. Global protective measures are urgently needed to prevent the extinction of these "living fossils."

Scorpions are increasingly threatened by habitat loss and degradation, which are caused by agriculture, mining, and urbanization and are exacerbated by climate change (4). Overharvesting for venom commerce and the exotic pet trade also puts scorpions at risk (5–7). Scorpion venom, valued at up to

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US\$10 million per liter, has attracted interest from pharmaceutical companies and amateur harvesters, fueling a surge in illegal “venom milking” farms (5). About 350 scorpion species are sold in the exotic pet trade (6, 7).

Despite these threats, only five scorpion species are listed by the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) (8), and until 2024, the conservation status of only three species (*Afrotychus braueri*, *Chiromachus ochropus*, and *Isometrus deharvengi*) had been assessed by the International Union for the Conservation of Nature (IUCN) (9). Twenty-six species were added to the IUCN Red List in 2025, mostly rated Least Concern (9), but those species account for less than 1% of known scorpion diversity. Furthermore, national red lists are not directly recognized by the IUCN, which hinders assessments of the actual number of threatened species globally.

Scorpions offer a glimpse into the evolution of life in deep time, and scorpion venom holds vast pharmacological potential, including applications for treating cancer and autoimmune diseases (10). The extinction of scorpions would erode ecosystem integrity and remove potential sources of human advancement. To understand the impact of threats and implement effective conservation of scorpions, National Biodiversity Strategies and Action Plans, which are designed to meet the goals of the Convention on Biological Diversity (CBD), should assess the status of at least 10% of the world’s scorpion species to align with the CBD’s 2030 targets (11, 12). In addition, global conservation bodies should add more scorpion species to the IUCN Red List and include threatened scorpion species in conservation management plans. Including more threatened scorpion species on the CITES lists, implementing rigorous regulation of the scorpion trade, and involving local communities in regional conservation programs would curb illegal venom extraction and species trafficking, thereby reducing pressure on wild populations.

Javier Blasco-Aróstegui^{1,2}, Jairo Moreno-González², Stephanie F. Loria³, Leonardo S. Carvalho⁴, Tharina L. Bird^{5,6}, Ahn D. Nguyen⁷, Hossein Barahoei⁸, Charles R. Haddad⁹, Lorenzo Prendini²

¹Centre for Ecology, Evolution and Environmental Changes, Departamento de Biologia Animal, Faculdade de Ciências, Universidade de Lisboa, Lisbon, Portugal. ²Arachnology Lab and Scorpion Systematics Research Group, Division of Invertebrate Zoology, American Museum of Natural History, New York, NY, USA. ³Department of Biology, College of Mathematics and Science, University of Central Oklahoma, Edmond, OK, USA. ⁴Campus Amílcar Ferreira Sobral, Universidade Federal do Piauí, Floriano, PI, Brazil. ⁵Ditsong National Museum of Natural History, Pretoria, South Africa. ⁶Department of Zoology and Entomology, University of Pretoria, Pretoria, South Africa. ⁷Institute of Ecology and Biological Resources, Vietnam Academy of Science and Technology, Cau Giay, Hanoi, Vietnam. ⁸Department of Agronomy and Plant Breeding, Agriculture Institute, Research Institute of Zabol, Zabol, Iran. ⁹Department of Zoology and Entomology, University of the Free State, Bloemfontein, South Africa. Email: lorenzo@amnh.org

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